UFC Fight Outcome Prediction Using Machine Learning

UE24CS352A - Machine Learning Mini-Project: Project Summary

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1. Problem Statement

Objective

The goal is to predict the outcome of a UFC fight before it takes place. For every fight the model must choose one of three outcomes: red corner wins, blue corner wins, or draw. This is a multi-class classification problem. The work follows the research paper "Applying Machine Learning Algorithms to Predict UFC Fight Outcomes" by McKinley McQuaide, which compares Generalized Linear Models, Multilayer Perceptrons, Decision Trees and Gradient Boosting on pre-fight fighter statistics.

Scope

We reproduce the paper's four model families, add Logistic Regression and Random Forest, and build a working predictor that takes the names of two fighters and returns the predicted result with probabilities. A useful predictor can support betting decisions and help coaches judge opponents.

2. Dataset Details

Source and size

We used the CSV file supplied with the research paper. It contains 2,318 fights from January 2014 to October 2018 and 894 columns. Each row is one fight. It holds details of both fighters (name, ID, age, height, weight, number of previous UFC fights), the winner, the method of victory, and round-by-round in-fight statistics for rounds 1 to 5 (strikes, takedowns, submission attempts, time in position). The file is smaller than the dataset described in the paper (3,355 fights, 1997 to 2019, 134 features), so absolute results are not directly comparable.

Data quality problems found

First, the in-fight statistics are only known after a fight ends, so using them directly would leak the answer. Second, every row has at least one missing value, so dropping incomplete rows as the paper did would leave no data. Third, 313 fights (13.5%) have no in-fight statistics at all. Fourth, the Date column mixes three formats and 501 row dates (21.6%) contradict the other fights of the same event. Fifth, 7.1% of the time-in-position values exceed the 300 seconds of a round, so they are unreliable. After removing 24 no-contest fights, 2,294 fights remain: red wins 57.8%, blue wins 41.5%, draws 0.7%.

3. Approach

Data cleaning

We removed no-contest fights, repaired each fight date using the median date of its event (this made fighter histories consistent in 99.5% of consecutive fights, against 89.0% for the raw dates), recovered missing age, height and weight from the same fighter's other fights, dropped location columns, and summed the round statistics into fight totals. Time-in-position columns were excluded. Fights without statistics keep missing values instead of zeros.

Leak-free feature engineering

The data was reshaped to one row per fighter per fight and sorted by date. For each fight, a fighter's features are computed only from his earlier fights: record and streaks, win rates by knockout, submission and decision, striking and takedown averages, accuracy, defence, damage absorbed, days since last fight, plus age, height, weight and previous UFC fights. This gives 38 features per fighter. The model receives the red features, the blue features and their differences, 114 features in total. A test on 10 random fights confirmed that no future information enters any feature.

Modelling and evaluation

Fights were split by time. The older 80% (1,831 fights, January 2014 to November 2017) trains the models and the newest 20% (463 fights, November 2017 to October 2018) tests them, which mirrors real use. Each model sits in a pipeline of median imputation, standardisation and classifier. Hyper-parameters were tuned by grid search with 5-fold time-series cross-validation on the training data only, and the final model was chosen by cross-validation accuracy, not by test accuracy. The baseline is always predicting the red corner.

4. Implementation Overview

Code structure

Everything is in one Python notebook (UFC_Fight_Prediction.ipynb) using pandas, NumPy, scikit-learn and matplotlib, submitted through a private GitHub repository that also contains a README with setup instructions and a requirements file. Part 1 (Member 1) covers the data audit, cleaning, exploratory analysis, feature engineering and the leakage test. Part 2 (Member 2) covers the time-based split, training and tuning of the six models, evaluation, final model and prediction function.

Prediction function

The function predict_fight(red, blue) matches fighter names even with spelling differences, builds both fighters' features from their full history, and returns the predicted winner with probabilities for red, blue and draw. Optional inputs set the fight date and override age or experience. The selected Random Forest is retrained on all 2,294 fights and saved with the fighter profiles. Only fighters present in the dataset can be predicted.

5. Conclusions

Results

Model	Cross-validation accuracy (training data)	Test accuracy (newest 463 fights)
Logistic Regression	57.0%	58.5%
Random Forest (selected model)	58.4%	57.5%
Gradient Boosting	57.5%	56.2%
Decision Tree (pruned)	55.9%	55.5%
Baseline: always predict red corner	Not applicable	55.5%
MLP Neural Network (L-BFGS)	54.6%	52.3%
SGD Classifier (Perceptron loss)	53.5%	51.0%

Random Forest was selected by cross-validation and scored 57.5% on the test set, about 2 points above the 55.5% baseline. With 463 test fights, its 95% interval (about 52.9% to 62.0%) still contains the baseline, so the gain is small. The paper shows the same pattern: its models reached about 58% to 61%, below its own 62.6% red-corner baseline.

Key findings

The age difference between the two fighters is the most important feature in both tree-based models, followed by each fighter's age, striking defence, layoff time and height, which agrees with the paper. Simple regularised models beat the MLP, which over-fits (about 80% training accuracy against 52% test accuracy). Draws (16 fights) are never predicted. The red corner is itself informative, so swapping corners changes the probabilities.

Limitations and future work

The data covers only 2014 to 2018, so career histories are truncated, and fighters outside the dataset cannot be predicted. Future work includes adding reach, stance and betting odds, Elo-style ratings, the full 1993 to 2019 history, and probability calibration.